

| Campaign                         | Manipulated                            | Held fixed                  | What it settles                                                    |
|----------------------------------|----------------------------------------|-----------------------------|--------------------------------------------------------------------|
| E1<br>concurrency                | feeds $N$ :<br>1, 9, 10, 12            | rate, host,<br>plan set     | the original question:<br>does broker choice matter?               |
| E-B<br>delay sweep               | injected delay<br>0-50 ms              | load, feeds                 | H1 effect size<br>(direction only; confounded)                     |
| E-A / E-A3<br>load sweep         | background load<br>0-12                | rate, feeds,<br>plan        | H2 utilization, H10 mixture,<br>$F_{\Delta}$ recovery              |
| E-A2<br>process count            | processes at<br>fixed rate             | aggregate<br>event rate     | H4 oversubscription                                                |
| E-C3 / E-C4<br>stamping          | ack stamp:<br>callback vs inline       | load, in-flight,<br>backend | H3 asymmetry<br>(replicated)                                       |
| window<br>sweep                  | observation window<br>60-600 s         | rate, host,<br>match        | start-up cost vs<br>per-event constant                             |
| transport<br>(x2)                | feeds, powered                         | verified<br>real-time rate  | broker transport,<br>equivalence + shift                           |
| E-A5/A5b/A7<br>stamping priority | SCHED_FIFO on<br>the stamping threads  | utilization,<br>to 0.003    | scheduling, not utilization<br>(8 pairs, 7-80x)                    |
| E-A6 / E-A6b<br>load geometry    | cores free vs<br>all duty-cycled       | achieved<br>utilization     | utilization is not the variable<br>(2.07x, 2.05x, at rho 0.7531)   |
| E-A9<br>run-queue trace          | nothing:<br>observation only           | load,<br>priority arm       | $P(\text{stall} > T_{\text{true}})$ predicts<br>the rate, unfitted |
| E-A10<br>transport sweep         | payload size;<br>$T_{\text{true}}$ 77x | load, hosts,<br>code path   | other side of the inequality;<br>tail index 0.34                   |
| E-A8<br>co-location              | broker on<br>the driver                | utilization,<br>to 0.003    | nothing: transport did not<br>move, so it is withheld              |
| OMB<br>external                  | instrumented<br>discard counter        | our broker,<br>under load   | the exposure is<br>not ours alone                                  |

*Every claim in the discussion traces to one row. No row both generates and tests the same hypothesis, and a row that settled nothing says so.*